

Analysis of Ground Current to Minimize the Potential Induced Degradation (PID) Effect in Solar PV Panels

MANISH KARN¹, K. RAMACHANDRA SEKHAR¹

¹Department of Electrical Engineering, Indian Institute of Technology Ropar, Rupnagar, Punjab 140001, India

Corresponding author: Manish Karn (e-mail: manishkarn.mk@gmail.com).

This work was carried out as part of an M.Tech thesis under the supervision of Assistant Professor Dr. K. Ramachandra Sekhar in the Department of Electrical Engineering, Indian Institute of Technology Ropar.

ABSTRACT Potential Induced Degradation (PID) is a long-term reliability concern for grid-connected photovoltaic (PV) systems, in which high system voltage drives a leakage current between the active solar cells and the grounded module frame. This paper analyzes the ground-current pathway responsible for PID and presents a hardware prototype for sensing this current on an operating PV system. The parasitic capacitance formed between a PV panel and earth is examined for a transformerless inverter topology, and a resonant-circuit model is used to relate this capacitance to the observable leakage-current oscillation period. A sensing circuit built around TL082 operational amplifiers, together with a three-phase IGBT-based pulse-width-modulated inverter, was designed, assembled, and tested on a 400 W outdoor PV array. The inverter was characterized across a range of modulation indices, and the sensing prototype was used to measure the common-mode voltage, and hence the ground current, between the module frame and earth during operation. The measured leakage current oscillates within a range consistent with parasitic-capacitance-driven behavior reported in the literature, supporting ground-current monitoring as a low-cost, field-deployable indicator of PID risk in grid-connected PV installations.

INDEX TERMS Ground current, leakage current, parasitic capacitance, photovoltaic module, potential induced degradation, solar energy, three-phase inverter.

I. INTRODUCTION

SOLAR photovoltaic (PV) generation is an increasingly important contributor to the global electricity mix, with grid-connected systems now dominating deployed capacity. PV modules are conventionally wired in series to build up string voltage, and, for safety, the metal module frames are grounded. As PV installations have grown, it has become clear that prolonged exposure to high voltage relative to ground can itself become a source of degradation. Panels operating at large potential differences with respect to earth experience Potential Induced Degradation (PID): a leakage current flows through the packaging materials of the module, and this current is associated with progressive, sometimes severe, loss of module power output. Because PID can develop silently before its effects become apparent in a module's current-voltage characteristic, it has emerged as one of the more troublesome long-term PV failure modes.

The severity of PID depends on the module technology, ambient temperature and humidity, and the system voltage relative to ground. As utility-scale arrays have moved toward

higher string voltages (up to 1500 V DC) to reduce ohmic losses, the high-voltage stress experienced by individual modules, and hence their exposure to PID, has increased. In transformerless inverter topologies, the parasitic capacitance between the PV panel and earth plays a central role: when this capacitance, the array, and the load or grid form a closed loop, the common-mode voltage produced by inverter switching drives a ground current through it. This current is undesirable not only because of its link to PID, but also because it distorts grid current, generates electromagnetic interference, and creates unsafe touch-voltage conditions at the module frame.

Although the relationship between PID and PV-to-ground parasitic capacitance is reasonably well understood theoretically, comparatively little published work reports a dedicated hardware system for directly sensing the ground current on a real, operating PV array. This paper addresses that gap. The main contributions are: (i) a resonant-circuit analysis relating the PV panel's parasitic capacitance to the measurable leakage-current oscillation period; (ii) design and con-

struction of a low-cost operational-amplifier-based sensing circuit for the frame-to-ground common-mode voltage; and (iii) experimental validation of the resulting prototype on an outdoor 400 W PV array feeding a purpose-built three-phase inverter, together with a characterization of the inverter's open-loop behavior across a range of modulation indices.

II. RELATED WORK

Brecl et al. [1] used electroluminescence imaging to show that high-voltage stress induces PID through ion migration between the solar cells and grounded module components, with severity governed by the cell-to-frame potential difference and accelerated by temperature and humidity. Voswinckel et al. [2] studied leakage currents beneath PV modules as a function of mounting arrangement, noting that the push toward 1000–1500 V DC systems (to reduce ohmic losses) increases exposure to diffusion-driven degradation. Del Cueto and McMahon [3] examined leakage currents under high-voltage bias in outdoor field conditions, correlating leakage current with meteorological conditions and cumulative leakage charge.

The PV-to-ground parasitic capacitance that sustains this leakage current has also been modeled extensively. Yu et al. [4] developed a parasitic-capacitance model accounting for rainwater on the module surface. Lopez et al. [5] showed that in a transformerless single-phase inverter, the ground capacitance, AC filter, and grid impedance form a resonant loop whose excitation depends on the PWM strategy: unipolar modulation introduces high-frequency ground-voltage content that can excite this resonance, while bipolar modulation keeps leakage current low. Wang et al. [6] proposed an on-line method for extracting the PV panel's parasitic capacitance C_{PV} from a single-phase full-bridge inverter, deriving a relationship between C_{PV} and the oscillation period of the leakage current; this resonant-circuit approach forms the analytical basis adopted in Section IV. Chen et al. [7] decomposed the panel-to-ground capacitance into cell-to-frame, cell-to-rack, and cell-to-ground components, reporting typical values of 7–220 nF/kW for glass-faced modules, rising sharply under humid conditions.

This body of work establishes both the mechanism by which high-voltage stress produces PID and the parasitic-capacitance pathway through which the associated leakage current flows, but comparatively little of it reports a hardware system for directly measuring this ground current on an installed, operating PV array. The present work is intended to help fill that gap.

III. PID MECHANISM

A standard PV module comprises five principal layers – front glass, front encapsulant, solar cells, rear encapsulant, and back sheet – enclosed within a grounded frame, as shown in Fig. 1. When the active cells are biased at high potential relative to the frame, leakage current can follow several paths through the front glass, encapsulant, and back sheet, illustrated for a conventional crystalline-silicon (c-Si) module in

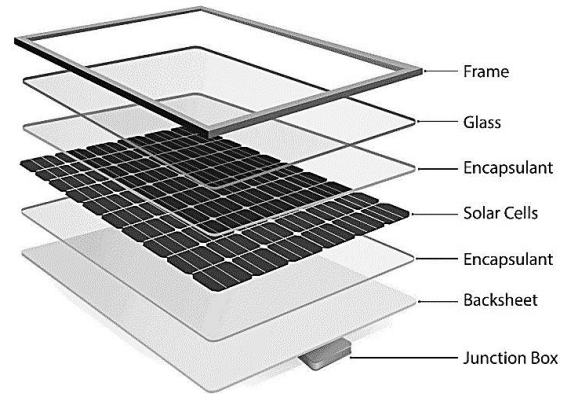


FIGURE 1. Exploded view of the principal layers comprising a PV module.

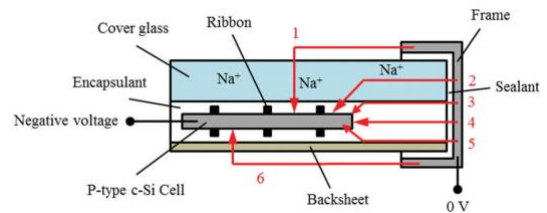


FIGURE 2. Cross-sectional view of a conventional c-Si PV module showing possible leakage-current pathways (1–6) from the cells to the grounded frame.

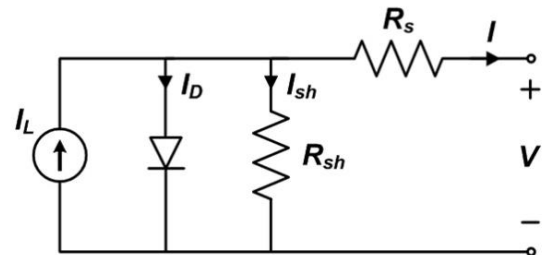


FIGURE 3. Single-diode equivalent circuit of a PV cell, including series resistance R_s and shunt resistance R_{sh} .

Fig. 2. Under negative bias, sodium (Na^+) ions originating from the soda-lime cover glass drift toward the cell surface and accumulate at the anti-reflective-coating/silicon interface, producing cell shunting; this pathway is typically the most severe under outdoor, humid conditions. PID therefore manifests most strongly in modules nearest the negative terminal of a series string, where the module-to-ground potential difference is largest.

PID reduces module performance by lowering the shunt resistance R_{sh} of the cell's single-diode equivalent circuit, shown in Fig. 3, which increases leakage current and visibly alters the module's I–V curve, as illustrated by the comparison between an affected and an unaffected module in Fig. 4. Detection methods for PID include I–V curve testing, electroluminescence imaging, infrared thermography, and statistical analysis of PV monitoring-portal data [12], [13].

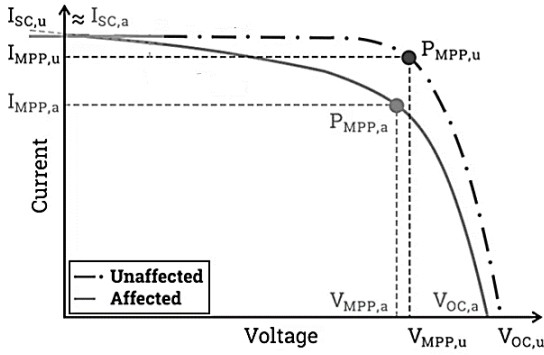


FIGURE 4. Comparison of I-V curves for a PID-affected module and an unaffected module.

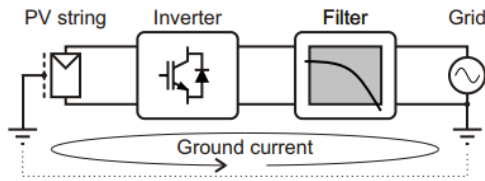


FIGURE 5. Formation of leakage/ground current in a PV-inverter system through the panel-to-ground parasitic capacitance.

IV. MATERIALS AND METHODS

A. FORMATION OF GROUND CURRENT AND RESONANT-CIRCUIT MODEL

In transformerless inverter topologies, the parasitic capacitance C_{PV} between the PV panel and ground must be considered whenever the panel frame is grounded. This capacitance, the PV array, and the load or grid together form a closed loop; switching-induced common-mode voltage drives a common-mode (ground) current through C_{PV} , illustrated in Fig. 5, resulting in grid-current distortion, electromagnetic interference, additional losses, and unsafe touch-voltage conditions [5].

Following [4], [6], [7], a non-isolated single-phase full-bridge inverter (Fig. 6) is used as the analytical vehicle for characterizing C_{PV} . Since $C_{PV} \gg \{C_{p1}, \dots, C_{p4}\}$ and the filter inductors $L_1, L_2 \gg$ the stray bus inductances, the high-frequency leakage loop reduces to the series resonant circuit of Fig. 7, with equivalent inductance $L_{eq} = L_1 L_2 / (L_1 + L_2)$ and resonant frequency

$$f_r = \frac{1}{2\pi\sqrt{L_{eq}C_{PV}}}. \quad (1)$$

Equating the resonant-circuit relation with the observed leakage-current oscillation period T gives

$$C_{PV} = \frac{L_1 L_2 T^2}{4\pi^2(L_1 + L_2)}, \quad (2)$$

so that C_{PV} can be estimated directly from a time-domain measurement of T , without a dedicated impedance test. As reported in [6] for a reference single-phase inverter platform

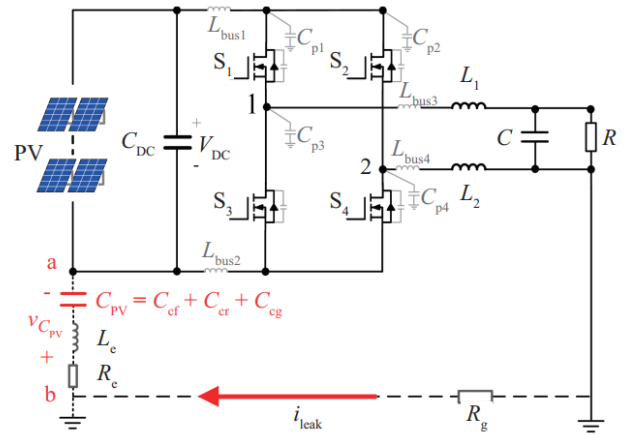


FIGURE 6. Single-phase full-bridge inverter with parasitic elements, used as the analytical vehicle for characterizing the PV-to-ground parasitic capacitance.

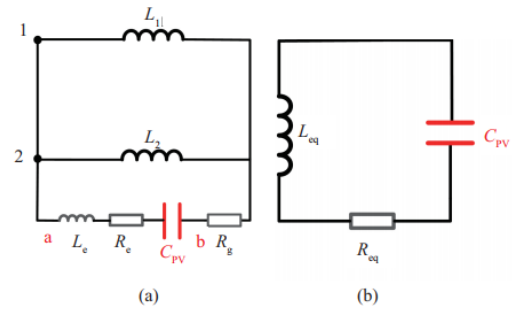


FIGURE 7. Equivalent resonant circuit model of the high-frequency leakage-current loop.

($V_{DC} = 18$ V, $L_1 = L_2 = 1$ mH, $C_{PV} = 5$ nF), the resulting leakage current oscillates between approximately -20 mA and $+20$ mA – a benchmark used for comparison in Section V.

B. GROUND-CURRENT SENSING PROTOTYPE

Using this relationship as a reference, a prototype was developed for sensing the leakage current flowing from a PV module's frame to ground. Fig. 8 shows the block diagram of the experimental setup: the PV module frame is connected to the sensing device and grounded in series with earth, while the three-phase resistive load is grounded at the same point. The connection between the load and the array generates a common-mode voltage that drives ground current through the panel's parasitic capacitance, which the sensing device measures.

The sensing device is built from TL082 operational amplifiers (selected for their 4 MHz gain-bandwidth product and 13 V/ μ s slew rate [14]), 1 k Ω /10 k Ω resistors, and a 9 V battery supply: a first stage configured as a unity-gain voltage follower presents high input impedance to the sensing node, followed by two gain-of-10 inverting-amplifier stages, giving an overall gain of approximately 100. The assembled hardware is shown in Fig. 9. On the bench, using a RIGOL

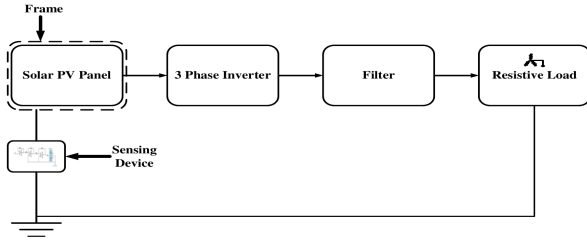


FIGURE 8. Block diagram of the complete experimental hardware setup.

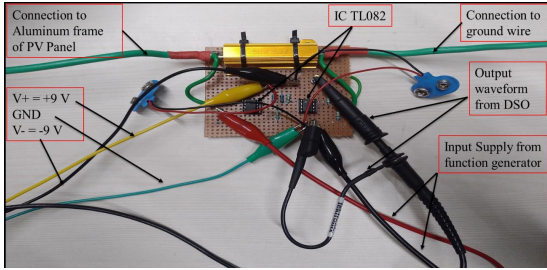


FIGURE 9. Assembled hardware for the ground-current sensing device, annotated with its connections to the PV module frame, the earth wire, the input supply from the function generator, and the output waveform to the DSO.

TABLE 1. Bench-Test Results for the Sensing Circuit at 5 kHz (Selected Points)

Input (mV)	Output (V)
40	3.96–4.00
20	2.00–2.08
10	1.08–1.12
5	0.56–0.60
1	0.188–0.212

DG1022Z function generator and a Tektronix TPS2024 oscilloscope, the circuit was verified to respond linearly across an input range of 1–40 mV at 5 kHz (Table 1), confirming a gain close to the design value of 100.

C. THREE-PHASE INVERTER AND PV ARRAY

A three-phase, two-level IGBT-based inverter (Fig. 10) was assembled using SKM75GB12T4 IGBT modules with SKYPER 32 PRO R gate drivers, driven by SPWM signals generated in MATLAB/Simulink and implemented on a C2000 Delfino LAUNCHXL-F28379D controller. The inverter output was passed through an LC filter ($L = 6$ mH, $C = 80$ μ F per phase) into a star-connected three-phase resistive load (16.2 Ω /phase). The complete assembled hardware, including gate drivers, adapter boards, DC bus bar, and controller, is shown in Fig. 11.

The DC-bus voltage requirement follows from $V_{L-L} = \sqrt{3}mV_{dc}/(2\sqrt{2})$; for a target line-to-line voltage of 230 V at unity modulation index, $V_{dc} = 2\sqrt{2} \times 230/(\sqrt{3} \times 1) = 375.58$ V. Based on the array's simulated maximum voltage of approximately 322 V at 1000 W/m² (Section V), a 450 V, 2200 μ F DC-link capacitor was selected, sized at 1.1 times the expected maximum bus voltage [15]. The experimental array, comprising eight series-connected 400 W modules

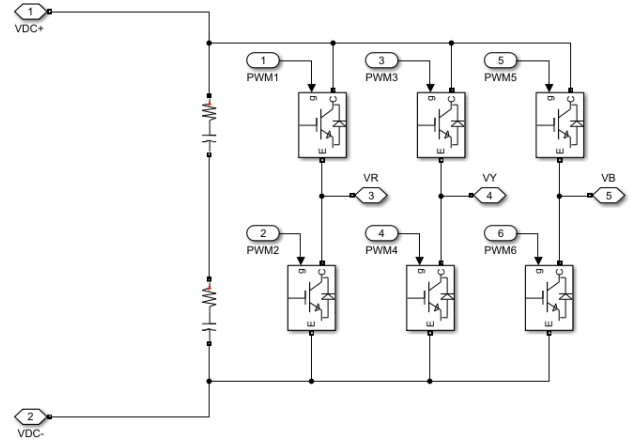


FIGURE 10. Three-phase, two-level IGBT inverter bridge with SPWM gating on all six switches.

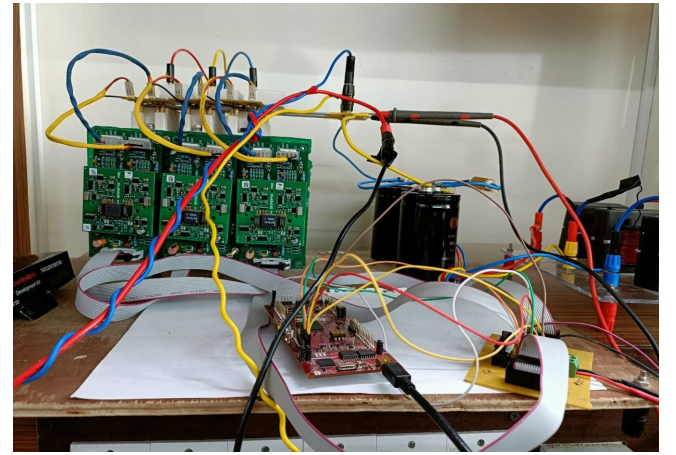


FIGURE 11. Complete assembled hardware for the three-phase inverter.

TABLE 2. Datasheet Parameters of the PV Module Used

Parameter	Value
P_{max} (W _p)	400 W
Voltage (V_{mp}) / Current (I_{mp})	40.25 V / 9.95 A
Open-circuit voltage (V_{OC})	48.45 V
Short-circuit current (I_{SC})	10.22 A
System voltage	1000 V DC

(Table 2), was installed outdoors and earthed together with the resistive load at a common earth pit, as shown in Fig. 12.

V. RESULTS AND DISCUSSION

Fig. 13 shows the simulated I–V and P–V characteristics of the eight-module array at 500, 750, and 1000 W/m². At 1000 W/m², the array reaches a maximum power point of approximately 3203.9 W at 322 V and 9.95 A, consistent with the DC-link sizing of Section IV.

With the array supplying $V_{DC} = 290$ V and $I_{DC} = 4.5$ A, the SPWM modulation index m was varied and the resulting three-phase voltage and current, after the LC filter, were recorded (Table 3); Fig. 14 shows a representative waveform



FIGURE 12. Installed solar PV array used for experimentation.

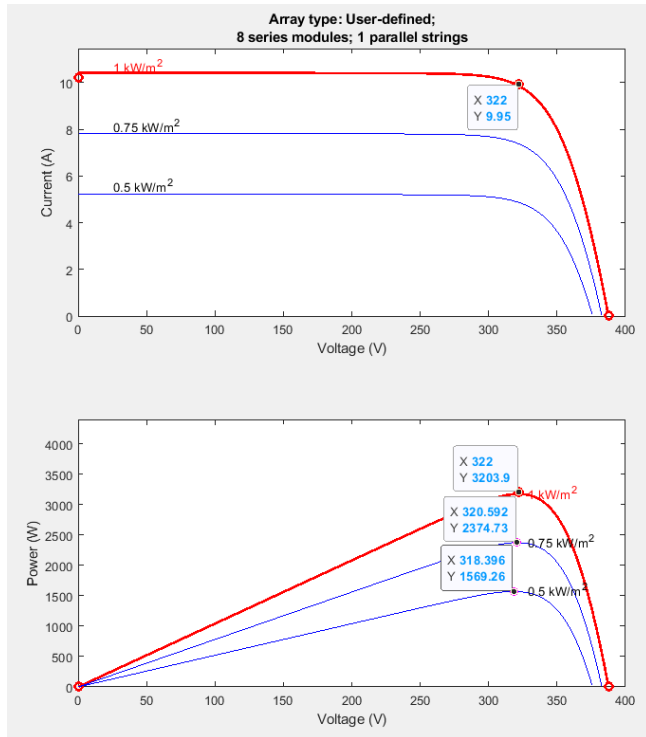


FIGURE 13. Simulated I–V and P–V curves of the eight-module experimental PV array at 500, 750, and 1000 W/m² irradiance.

TABLE 3. Three-Phase Voltage and Current vs. Modulation Index ($V_{DC} = 290$ V, $I_{DC} = 4.5$ A)

m	Voltage (V)	Current (A)
1.0	91.83	5.64
0.6	58.64	3.61
0.2	14.16	0.87

at $m = 1.0$. Both voltage and current scale approximately linearly with m , confirming stable, predictable open-loop operation of the assembled inverter.

With the sensing device connected between the PV array frame and earth, and the inverter driving the resistive load, the amplified common-mode voltage was recorded, as shown in Fig. 15. The measured peak-to-peak output is approximately 10.4 V, which, after removing the sensing circuit's gain of 100, corresponds to a frame-to-ground voltage of 0.104 V and a peak-to-peak leakage current of approximately

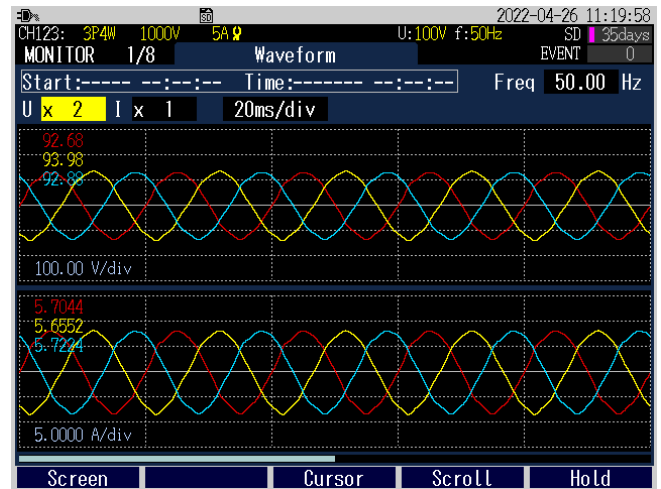


FIGURE 14. Three-phase voltage and current waveform at modulation index $m = 1.0$.

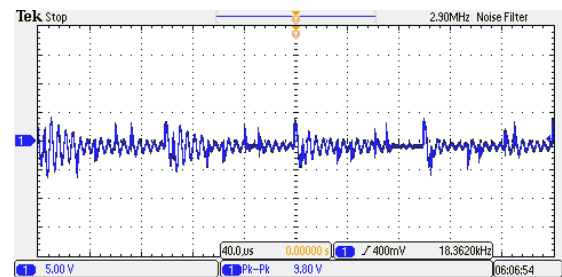


FIGURE 15. Amplified frame-to-ground common-mode voltage waveform captured through the sensing prototype, showing a dominant high-frequency component near 2.90 MHz.

0.052 A – that is, the ground current oscillates between roughly -26 mA and $+26$ mA. This is consistent in order of magnitude with the -20 mA to $+20$ mA leakage-current oscillation reported for the reference platform of [6] ($C_{PV} = 5$ nF), supporting the use of (2) as a physically consistent explanation for the observed behavior and confirming the sensing prototype as a viable, low-cost means of monitoring PID-related ground current without a dedicated impedance-analyzer measurement.

VI. CONCLUSION AND FUTURE WORK

This paper analyzed the ground current responsible for Potential Induced Degradation in solar PV panels and presented a hardware prototype for sensing this current on an operating PV system. A resonant-circuit model was used to relate the PV-to-ground parasitic capacitance to the observable leakage-current oscillation period, motivating a TL082-based sensing circuit with a gain of approximately 100. Validated on the bench and on a 400 W outdoor PV array feeding a three-phase IGBT inverter, the prototype measured a ground current oscillating between approximately -26 mA and $+26$ mA, consistent with parasitic-capacitance-driven behavior reported in the literature, while the inverter itself exhibited stable, linearly-scaling output across modulation

indices from 0.1 to 1.0. Future work will extend this analysis to closed-loop, grid-interfaced inverter operation, compare the ground-current profile under SPWM and SVPWM modulation, and examine the sensing approach for NPC and dual-inverter topologies incorporating MPPT.

ACKNOWLEDGMENT

The authors thank the Renewable Power Modulation (RPM) Laboratory, Department of Electrical Engineering, Indian Institute of Technology Ropar, for laboratory access and technical support.

REFERENCES

- [1] K. Brecl, M. Bokalič, and M. Topič, "Examination of PV silicon module degradation under high voltage bias and damp heat by electroluminescence," *J. Sol. Energy Eng.*, vol. 139, no. 3, Jun. 2017.
- [2] S. Voswinckel, P. Manz, C. Schmidt, and V. Wesselak, "Investigation of leakage currents depending on the mounting situation in accordance to amorphous silicon modules," *Energy Procedia*, vol. 57, pp. 56–64, 2014.
- [3] J. A. del Cueto and T. J. McMahon, "Analysis of leakage currents in photovoltaic modules under high-voltage bias in the field," *Prog. Photovolt., Res. Appl.*, vol. 10, no. 1, pp. 15–28, 2002.
- [4] S. Yu, J. Wang, X. Zhang, and F. Li, "Complete parasitic capacitance model of photovoltaic panel considering the rain water," *Chin. J. Electr. Eng.*, vol. 3, no. 2, pp. 77–84, 2017.
- [5] O. Lopez, R. Teodorescu, F. Freijedo, and J. Doval-Gandoy, "Eliminating ground current in a transformerless photovoltaic application," in *Proc. IEEE Power Eng. Soc. Gen. Meeting*, Jun. 2007, pp. 1–5.
- [6] J. Wang, Y. Xun, X. Liu, and S. Yu, "An on-line extraction method for the parasitic capacitance of the photovoltaic panel," *CPSS Trans. Power Electron. Appl.*, vol. 4, no. 4, pp. 320–327, Dec. 2019.
- [7] W. Chen, X. Yang, W. Zhang, and X. Song, "Leakage current calculation for PV inverter system based on a parasitic capacitor model," *IEEE Trans. Power Electron.*, vol. 31, no. 12, pp. 8205–8217, Dec. 2016.
- [8] G. M. Masters, *Renewable and Efficient Electric Power Systems*. Hoboken, NJ, USA: Wiley, 2004.
- [9] S. Bowden and C. Honsberg, "Photovoltaics education," Solar Power Labs, Arizona State Univ., Tempe, AZ, USA. [Online]. Available: <https://www.pveducation.org/>
- [10] "Alternative energy tutorials." [Online]. Available: <https://www.alternative-energy-tutorials.com/photovoltaics/photovoltaics.html>
- [11] K. Vidyandandan, "An overview of factors affecting the performance of solar PV systems," *Energy Scan*, NTPC Ltd. Corporate Planning House J., no. 27, pp. 2–8, Feb. 2017.
- [12] W. Luo et al., "Potential-induced degradation in photovoltaic modules: A critical review," *Energy Environ. Sci.*, vol. 10, no. 1, pp. 43–68, Jan. 2016.
- [13] D. D. de Rooij, "Potential induced degradation (PID): How to reverse or prevent solar PID?" *Sinovoltaics*. [Online]. Available: <https://sinovoltaics.com/technology/potential-induced-degradation-pid-how-to-reverse-prevent/>
- [14] TL082 Wide Bandwidth Dual JFET Input Operational Amplifier, Texas Instruments, Dallas, TX, USA.
- [15] J. Sylvestre, "Inverter DC link capacitor selection," *Specter Eng.* [Online]. Available: <https://www.specterengineering.com/blog/2019/9/7/dc-link-capacitor-selection-for-your-inverter>

• • •